

## ChatGPT

Share

give me the datasheet for HR-053A

I'm checking what **HR-053A** refers to first, then I'll turn that into a clean datasheet-style summary with the key specs and likely manufacturer details.

Thought for 24s

I couldn't reliably identify **HR-053A** from web sources alone.

The problem is that "**HR-053A**" appears too ambiguous and does not point to a single clearly identifiable product or official manufacturer datasheet in the search results. The results I found were unrelated or conflicting, and I did **not** find a trustworthy official datasheet tied to that exact model number. IDOT Apps +2

To give you the correct datasheet, please send **one** of these:

- the **product name / category**
- the **manufacturer / brand**
- a **photo of the label**
- a **product link**

Once you send that, I can turn it into a clean datasheet-style sheet right away.

 Sources

how about RFID chips like U8/U9/R6/R6P?

I'm checking the exact chip family names and official spec differences so I can give you a clean comparison instead of a vague summary.

Thought for 13s

Yes — **U8 / U9 / R6 / R6P** usually refer to **UHF RAIN RFID tag chips**, not complete readers or cards.

What they are

- U8 usually means NXP UCODE 8
- U9 usually means NXP UCODE 9
- R6 usually means Impinj Monza R6
- R6P usually means Impinj Monza R6-P Impinj +2

These are **passive UHF RFID ICs** used inside labels, stickers, hang tags, hard tags, and asset tags. They are designed for **RAIN RFID / EPC Gen2 / ISO 18000-63** type systems in the **860–960 MHz** band. NXP +1

Simple comparison

| Chip             | Vendor | Typical positioning                               | Notable point                                                      |
|------------------|--------|---------------------------------------------------|--------------------------------------------------------------------|
| UCODE 8 (U8)     | NXP    | Older high-volume retail/logistics tag IC         | Established design, widely used                                    |
| UCODE 9 (U9)     | NXP    | Newer generation than U8                          | Better performance, drop-in migration path from U8 in many designs |
| Monza R6 (R6)    | Impinj | General-purpose high-performance tag IC           | Broad use in retail, logistics, healthcare                         |
| Monza R6-P (R6P) | Impinj | R6 plus extra protection/privacy-related features | More memory and password/privacy options                           |

Main differences

1. Vendor family

- U8/U9 belong to NXP’s UCODE family. NXP +2
- R6/R6P belong to Impinj’s Monza family. Impinj +1

2. Generation

- UCODE 9 is the newer generation compared with UCODE 8. NXP explicitly says UCODE 9 can reuse existing UCODE 8 antenna designs as an easier migration path. NXP +1
- Monza R6 is an older Impinj generation and Impinj notes that Monza R6 is in the end-of-life process. Impinj

### 3. Memory / protection features

- R6-P adds extra features beyond basic R6, including **added user memory, password protection, and range reduction/privacy-related capability.** Impinj
- **UCODE 9** is positioned for demanding inventory and logistics applications; NXP also has related variants like **9xe** and **9xm** for larger EPC or user-memory needs. NXP +2

### 4. Status in the market

- **UCODE 8/8m** is listed by NXP as **No Longer Manufactured** on its product page. NXP
- **UCODE 9** is the newer active direction in that family. NXP +1
- **Monza R6** is older and moving toward EOL, so for new designs people often look at newer Impinj parts instead of starting fresh with R6. Impinj +1

### Where they are used

All four are commonly used in:

- retail apparel tags
- warehouse and logistics labels
- baggage tags
- asset tracking labels
- item-level inventory tags Impinj +2

### Practical buying/design meaning

When a tag seller says:

- “**U8 label**” → usually a label using **NXP UCODE 8**
- “**U9 label**” → usually **NXP UCODE 9**
- “**R6 label**” → usually **Impinj Monza R6**
- “**R6P label**” → usually **Impinj Monza R6-P** Impinj +2

But that still does **not** fully define the tag performance by itself, because actual performance also depends heavily on:

- antenna design
- tag size
- material the tag is attached to
- mounting environment
- reader power and antenna setup Impinj +2

### Recommendation

For new passive UHF RFID label selection:

- **U9** is generally the more modern NXP choice than **U8**. NXP +2
- **R6P** is chosen over plain **R6** when you need extra protection/privacy-style features.

Impinj

- For fresh designs, I would treat **U8** and **R6** more as legacy/older references unless a specific tag/inlay is already qualified in your system. Impinj +1

I can also make you a **clean datasheet-style comparison sheet for U8 vs U9 vs R6 vs R6P** with memory, sensitivity, protocol, use cases, and selection advice.



yes, please

I'm pulling the official chip datasheets and product pages so the comparison is based on manufacturer specs, not reseller summaries.

Thought for 32s

Below is a **clean datasheet-style comparison sheet** for **U8 / U9 / R6 / R6P**.

## UHF RFID Chip Comparison Datasheet

### 1. Overview

These part names usually refer to **passive UHF RAIN RFID tag chips** used inside labels, tags, and inlays:

- **U8** = NXP UCODE 8
- **U9** = NXP UCODE 9
- **R6** = Impinj Monza R6
- **R6P** = Impinj Monza R6-P NXP +3

All four are intended for **UHF RFID / RAIN RFID** operation in the **860–960 MHz** class and are used for applications such as retail, logistics, baggage, healthcare, and inventory tracking. NXP +3

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### 2. Quick Comparison Table

| Item              | U8                                     | U9                                            | R6                                     | R6P                                     |
|-------------------|----------------------------------------|-----------------------------------------------|----------------------------------------|-----------------------------------------|
| Vendor            | NXP                                    | NXP                                           | Impinj                                 | Impinj                                  |
| Product family    | UCODE 8                                | UCODE 9                                       | Monza R6                               | Monza R6-P                              |
| Protocol          | EPC Gen2v2 / EPCglobal v2.0.1          | EPC Gen2v2.1 / EPCglobal v2.1                 | RAIN RFID / ISO 18000-63 / EPC Gen2v2  | RAIN RFID / ISO 18000-63 / EPC Gen2v2   |
| Read sensitivity  | -23 dBm                                | -24 dBm                                       | -22 dBm                                | -22 dBm                                 |
| Write sensitivity | -18 dBm                                | -22 dBm                                       | -17 dBm                                | -17 dBm                                 |
| EPC memory        | 128-bit                                | 96-bit                                        | 96-bit                                 | 96 / 128-bit                            |
| TID               | 96-bit                                 | 96-bit                                        | serialized Tag ID supported            | serialized Tag ID supported             |
| User memory       | 0-bit on U8                            | none stated in base U9                        | None                                   | 32 / 64-bit                             |
| Passwords         | 32-bit Access + 32-bit Kill            | 32-bit Kill                                   | None listed as data-protection feature | Access/Kill                             |
| Temperature       | -40 °C to +85 °C                       | -40 °C to +85 °C                              | not shown in the spec table I checked  | not shown in the spec table I checked   |
| Special notes     | Untraceable, PSF/EAS, Brand Identifier | better sensitivity, U8 antenna migration path | EOL process                            | privacy / protection / short-range mode |

Sources: NXP UCODE 8/8m datasheet and page; NXP UCODE 9 datasheet and page; Impinj Monza R6 series spec table and product page. NXP +8

### 3. Individual Datasheet-Style Summaries

#### U8 — NXP UCODE 8

**Type:** Passive UHF RFID IC

**Vendor:** NXP

**Frequency class:** 860–960 MHz air interface class

**Protocol:** EPCglobal Gen2 v2.0.1 / UHF RFID air interface at 860–960 MHz

**Read sensitivity:** about -23 dBm

**Write sensitivity:** about -18 dBm

**Memory:**

- Reserved: **64-bit** total for **32-bit Access** + **32-bit Kill**
- EPC: **128-bit**
- TID: **96-bit**
- User memory: **0-bit**

**Special features:** Self Adjust, Memory Safeguard, Brand Identifier, Untraceable, pre-serialization of 96-bit EPC, Product Status Flag for EAS use

**Operating temperature:** -40 °C to +85 °C

**Status:** NXP lists UCODE 8/8m product page as **No Longer Manufactured**. NXP +4

#### U9 — NXP UCODE 9

**Type:** Passive UHF RFID IC

**Vendor:** NXP

**Frequency class:** 860–960 MHz air interface class

**Protocol:** EPCglobal Gen2 v2.1

**Read sensitivity:** -24 dBm

**Write sensitivity:** -22 dBm

**Memory:**

- Reserved: **32-bit Kill**
- EPC: **96-bit**
- TID: **96-bit**
- User memory: not listed in the base UCODE 9 memory table I checked

**Special features:** Self-Adjust, Memory Safeguard, Dynamic backscatter, pre-serialized 96-bit EPC

**Operating temperature:** -40 °C to +85 °C

**Migration note:** NXP says existing UCODE 8 inlay antenna designs can be reused, making U9 an easier upgrade path from U8. NXP +3

## R6 — Impinj Monza R6

**Type:** Passive UHF RAIN RFID endpoint IC

**Vendor:** Impinj

**Protocol:** RAIN RFID / ISO 18000-63 / EPCglobal Gen2v2 compliant

**Read sensitivity:** -22 dBm

**Write sensitivity:** -17 dBm

**Memory:**

- EPC: 96-bit
- User memory: None

**Special features:** AutoTune, Integra, Enduro, TagFocus, FastID

**Use cases:** retail, healthcare, hospitality, supply chain, baggage, general inventory

**Lifecycle note:** Impinj states **Monza R6 is in the end-of-life process.** [Impinj +1](#)

## R6P — Impinj Monza R6-P

**Type:** Passive UHF RAIN RFID endpoint IC

**Vendor:** Impinj

**Protocol:** RAIN RFID / ISO 18000-63 / EPCglobal Gen2v2 compliant

**Read sensitivity:** -22 dBm

**Write sensitivity:** -17 dBm

**Memory:**

- EPC: 96 / 128-bit
- User memory: 32 / 64-bit

**Protection / privacy:**

- Access/Kill
- short-range mode / range reduction capability

**Special features:** AutoTune, Integra, Enduro, TagFocus, FastID

**Typical positioning:** better for loss prevention, brand protection, and enhanced privacy than plain R6. [Impinj +1](#)

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## 4. Practical Selection Guide

### Choose U8

Use it mainly when you already have a legacy inlay or approved design based on U8. For new designs, it is harder to recommend because NXP marks UCODE 8/8m as **No Longer Manufactured.** [NXP](#)

## Choose U9

This is the strongest general-purpose choice among the NXP options here when you want a newer chip with **better sensitivity** and easier migration from U8 antenna designs.

NXP +1

## Choose R6

Reasonable only when a current tag or inlay you already buy is qualified around it. For fresh designs, note that Impinj says Monza R6 is **in the EOL process**.

Impinj +1

## Choose R6P

Best when you need **extra control**, such as user memory, access/kill protection, or privacy-oriented reduced range behavior.

Impinj +1

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## 5. Bottom-Line Recommendation

For a new general passive UHF label/inlay design:

- Best modern NXP direction: U9
- Best Impinj option of these four when privacy/protection matters: R6P
- Legacy / older references: U8 and R6

NXP +2

## 6. Important Note

The chip alone does not determine final read range. Real performance depends heavily on the antenna/inlay design, tag size, mounting material, item contents, and the reader/antenna setup.

NXP +2

I can turn this into a **clean one-page PDF-style datasheet sheet** next.



give me the prices for it

I'm checking current market listings for each chip family so I can give you a usable price range rather than a single unreliable number.



I found usable current pricing for **tags/inlays built with these chips**. Plain **chip-only** pricing is much harder to verify publicly, especially for older parts like U8 and R6.

Thought for 41s

I found **current public prices for tags/inlays that use those chips**, not clean distributor pricing for the **bare silicon IC alone**. For these RFID families, public web pricing is much more commonly shown for finished **labels, wet inlays, cable tags, or paper tags** than for the chip by itself. ABC RFID +3

Current price snapshot

| Chip family | Example current public price found                                                                                                                                                                                      | Notes                                                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| U8          | <b>\$0.40–\$0.55 each</b> for an ABC RFID cable tag, depending on quantity; <b>\$0.77 each at qty 10</b> and <b>\$0.72 each at qty 100</b> for a GoToTags on-metal label                                                | These are finished tags, not bare chips. <small>ABC RFID +1</small>                                                                                     |
| U9          | <b>\$0.07–\$0.20 each</b> for an ABC RFID wet inlay, depending on quantity; another seller shows <b>\$96–\$449 per 1,000–5,000 pcs</b> , which works out to about <b>\$0.096–\$0.0898 each</b>                          | U9 public pricing is easiest to find on standard wet inlays. <small>ABC RFID +1</small>                                                                 |
| R6          | <b>\$0.13 each</b> for a Smartrac paper tag on one site; another current listing shows <b>\$9.89 for 20 pcs</b> , about <b>\$0.49 each</b> ; GoToTags also shows a Monza R6 paper tag at <b>\$1.00 each for qty 10+</b> | R6 pricing is scattered, likely because it is older and listings are inconsistent by form factor and stock status. <small>rfdome.com - rf... +2</small> |
| R6P         | <b>\$0.07–\$0.20 each</b> for ABC RFID wet inlays, depending on quantity; GoToTags shows <b>\$0.21 each at qty 2,000</b> and “as low as <b>\$0.14</b> ” on its UHF tags page                                            | R6-P has the clearest currently available public tag pricing. <small>ABC RFID +2</small>                                                                |

Practical estimate

For **standard passive UHF labels/inlays** built on these chip families, the public market prices I found suggest these rough working ranges:

- **U8:** about **\$0.40 to \$0.77 each** in the examples I found. ABC RFID +1
- **U9:** about **\$0.07 to \$0.20 each**, with bulk listings clustering around **\$0.09–\$0.10 each**.

ABC RFID +1

- **R6:** about **\$0.13 to \$1.00 each** in current public listings, with wide variation by seller and product style. rfdome.com - rf... +2
- **R6P:** about **\$0.07 to \$0.21 each** in the clearest current listings I found. ABC RFID +2

## Best way to interpret this

These numbers are **not directly comparable as pure chip prices**, because each listing is for a different finished product type: cable tag, wet inlay, paper label, or on-metal label. Form factor, antenna size, adhesive, print surface, packaging, and order quantity all move the price a lot. ABC RFID +3

If your real goal is **procurement**, the most realistic quick-budget numbers from the public listings are:

- **Basic U9 or R6P wet inlays:** budget around **\$0.07–\$0.10 each** in higher volume. ABC RFID +1
- **Specialized U8 tags** such as cable tags or on-metal labels: budget around **\$0.40–\$0.77 each** from the examples found. ABC RFID +1
- **Older R6 products:** pricing is less consistent, so I would treat it as **stock-dependent**.

rfdome.com - rf... +1

I can also build a **clean price comparison sheet** for **U8 vs U9 vs R6 vs R6P** with per-piece price, product type, quantity break, and source.

